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Problem solving protocol

Systematic analysis of supervised machine learning as an effective approach to predicate β-lactam resistance phenotype in *Streptococcus pneumoniae*

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Abstract

Streptococcus pneumoniae is the most common human respiratory pathogen, and β-lactam antibiotics have been employed to treat infections caused by *S. pneumoniae* for decades. β-lactam resistance is steadily increasing in pneumococci and is mainly associated with the alteration in penicillin-binding proteins (PBPs) that reduce binding affinity of antibiotics to PBPs. However, the high variability of PBPs in clinical isolates and their mosaic gene structure hamper the predication of resistance level according to the PBP gene sequences. In this study, we developed a systematic strategy for applying supervised machine learning to predict *S. pneumoniae* antimicrobial susceptibility to β-lactam antibiotics. We combined published PBP sequences with minimum inhibitory concentration (MIC) values as labelled data and the sequences from NCBI database without MIC values as unlabelled data to develop an approach, using only a fragment from *pbp2x* (750 bp) and a fragment from *pbp2b* (750 bp) to predicate the cefuroxime and amoxicillin resistance. We further validated the

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performance of the supervised learning model by constructing mutants containing the randomly selected *pbps* and testing more clinical strains isolated from Chinese hospital. In addition, we established the association between resistance phenotypes and serotypes and sequence type of *S. pneumoniae* using our approach, which facilitate the understanding of the worldwide epidemiology of *S. pneumonia*.

Key words: *Streptococcus pneumonia*; β-lactam resistance; penicillin-binding protein; genotypic antimicrobial susceptibility testing; supervised machine learning

Introduction

The widespread dissemination of antimicrobial resistance genes among pathogens poses a clinical super challenge [1]. In 2014, antimicrobial-resistant infections were estimated to take the lives of at least 700 000 every year globally, and it is expected that the rise of resistance would lead to 10 million people dying by the year 2050 [2]. To combat antimicrobial resistance, antimicrobial stewardship is urgently to be improved, which calls for the development of rapid antimicrobial susceptibility testing (AST). Rapid AST could guide proper usage of antibiotics and reduce the evolution of multiple drug-resistant strains [3]. Currently, classical phenotypic AST methods based on cultivation of the target pathogens are the gold standard. However, these methods usually take several days including sample processing, which often defer immediate treatment decisions [4]. Genotypic AST has now occupied an increasingly important diagnostic niche, which aims to provide rapid AST results (1–3 h) by polymerase chain reaction (PCR) mediated detection of genetic markers of resistance phenotypes [5]. In addition, whole-genome sequencing (WGS) of cultured microorganisms or direct metagenomic sequencing of clinical materials have been proposed as next generation genotypic AST due to the rapid development of sequencing technology and reduction of sequencing cost [6]. However, the drawback of genotypic AST is imperfect correlation between presence of antibiotic resistance genes (ARGs) or mutations with their phenotypes.

Streptococcus pneumoniae is the most common human respiratory pathogen, causing mainly pneumonia, acute otitis media and bacterial meningitis [7]. β-lactam antibiotics have been applied to treat infection caused by *S. pneumoniae* for decades [8]. However, the incidence of penicillin-non-susceptible pneumococci has markedly escalated worldwide, and the ratio has been over 20% in some countries [9]. β-lactam resistance in pneumococci is mainly associated with the alteration in penicillin-binding proteins (PBPs) that reduce binding affinity of antibiotics to PBPs [10]. *Streptococcus pneumoniae* genome encodes six PBPs, and among which PBP2x, PBP2b and PBP1a are the major players in the β-lactam resistance. PBP2x and PBP1a contribute to cephalosporins resistance, while PBP1a and PBP2b are involved in penicillin resistance [11]. These three PBPs are highly variable in clinical isolates, and their mosaic gene structure indicates existence of interspecies recombination events and adaptive evolution of *pbp1a* (gene encoding PBP1a), *pbp2x* (gene encoding PBP2x) and *pbp2b* (gene encoding PBP2b) [11]. Therefore, it remains to be a great challenge to predict the antimicrobial resistant level of clinical isolates according to the sequences of PBP genes.

System biology approaches have been intensively employed to investigate bacterial infections, which provide novel insight into bacterial adaptive evolution, host-pathogen interactions as well as the connections between infections and cancer [12, 13]. Various machine learning (ML) methods have been used to discover knowledge as well as make predictions or decisions from complex biological and medical datasets. Unsupervised learning such as independent component analysis has been applied to extract gene expression features and cluster genes from microarray gene expression data [13] and temporal RNA sequencing data [14]. Supervised learning (SL) is a category of ML approach that predicts certain properties of objects from their features after training from existing samples of properties and features [15] while knowledge of exact mechanisms between properties and features are not necessarily needed. Multi-omics information, including mutations, gene expression hubness, known regulatory roles, copy number variation and gene methylation were incorporated in an ML model to identify robust molecular markers for targeted treatment of acute myeloid leukaemia [16]. A Support Vector Machine-based ML method was used to predicate mycoplasma hominis proteins targeting in mitochondria and cytoplasm of host cells and expose their implications in prostate cancer etiology [17, 18]. ML has also been used for prediction of antimicrobial resistance from genomes. Set Covering Machine-based ML models were constructed to predict antibiotic resistance phenotypes and identify biomarker using reference-free genome comparisons [19]. This approach was also used to identify PBP transpeptidase signatures for tracking and predicting β-lactam resistance levels in *S. pneumoniae* by application of mode model and random forest (RF) model [20]. However, ML model encounters several problems in biological datasets. The uncertainty of biological datasets is usually larger than datasets from engineering fields. Sequence data with labelled target antimicrobial resistance values that we want to predict is expensive and slow to acquire, while the amounts of accumulated unlabelled data that don't have antimicrobial resistance values in public database such as NCBI are continuously increasing, which might be exploited to enhance performance of models based on labelled target values. Furthermore, when applying ML model in practice, such as for rapid AST, features should be thoroughly studied and selected, which can reduce the time and cost to acquire and still maintain satisfying model performance.

In this study, we developed a systematic strategy for applying SL to predict AST of β-lactam antibiotic resistance. We firstly designed a model evaluation metric considering the inaccuracy of measured minimum inhibitory concentration (MIC) values and built four ML models to show the robustness of ML to predict *S. pneumoniae* resistance. Next we identified highly variant amino acid (AA) loci (HVLs) from unlabelled strains in NCBI database and showed that these HVLs were sufficient to predict antibiotic resistance, where the variance of an AA loci was measured by categorical variance. Strong associations existed between HVLs and fragments from each sequence can be used to predict antibiotic resistance with good accuracy, which allows one Sanger sequencing reaction to predicate the resistance phenotype. The *pbp* genes of randomly selected strains from NCBI database predicted as resistant and susceptible by our models were transformed into susceptible strains R6, and PBPs from predicated resistant strains significantly increased resistant level of the

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receipt strain and PBPs from predicated susceptible strains didn't alter the phenotype of the receipt strain. Furthermore, we validated the performance of the SL model with our collection of *S. pneumoniae* from a local hospital. Finally, we predicate more than 8000 *S. pneumoniae* strains available in NCBI database and provided a correlation between resistant phenotype and serotype and phylogenesis. Overall, our study established an effective genotypic AST approach for detecting β-lactam resistance levels in *S. pneumoniae*.

Materials and Methods

Datasets

We used available data set that contained 4300 pneumococcal strains with data of transpeptidase domain (TPD) sequences and labelled MIC of antibiotics, which Li et al. have produced [20, 21]. A total of 6748 and 8238 *S. pneumoniae* genomes were downloaded from NCBI Reference Sequence Database [https://www.ncbi.nlm.nih.gov/refseq/] in 2017 and 2018, separately. The sequences of *pbp1a*, *pbp2b* and *pbp2x* genes of *S. pneumoniae* strains were extracted from NCBI genome database.

Categorical variance

Categorical variance provides a measurement of variability for categorical random variable [22]. In this work categorical variance was used to quantify the variabilities of PBP AA loci by comparing different strains. In N *S. pneumoniae* strains, there are m_i possible AAs at locus i. The AA of k_j strains is j ($1 \leq j \leq m_i$) at locus i. Categorical variance of locus i (v_i) is shown in equation 1.

$$v_i = 1 - \sum_{j=1}^{m_i} \frac{k_j^2}{N^2} \tag{1}$$

Statistical analysis

Cramer's V was calculated using python scipy 0.19.1 [23]. The coefficient of determination,which describes the correlation between two variables, was used to estimate the contribution of AA loci to the variance of cefuroxime (CFX) or amoxicillin (AMO) MIC values, gathered by Li et al. from the Active Bacterial Core surveillance [24]. The coefficient of determination of locus i to MIC value R_i^2 is described in equation 2.

$$R_i^2 = 1 - \frac{\sum_{S=1}^{N} \left(MIC_s - \overline{MIC}_{is}\right)^2}{\sum_{S=1}^{N} \left(MIC_s - \overline{MIC}\right)^2} \tag{2}$$

Where N is the total number of strains; MIC_s is the MIC of strains; $\overline{MIC}$ is the mean MIC of all strains; and $\overline{MIC}_{is}$ is the mean MIC of strains whose AA at locus i is the same. Analysis of Variance (ANOVA) method tests whether the population means of several groups are equal. The *P*-value calculates the probability one can get the existing results if the hypothesis that the group means are equal is true. Usually, ANOVA test cannot differentiate the group means if *P*-value is greater than 0.05 [25]. In this study we use ANOVA to test whether different models or sets of features have significant differences in the performance scores. *P*-values of ANOVA were calculated using python scipy 0.19.1 [23].

Machine learning models

Four regression models were used to predict MIC values from AA sequences: RF, K nearest neighbours (KNN), support vector machine (SVM) and artificial neural network (ANN). One RF classifier model was used to predict remaining HVLs using sequence fragment. RF, KNN and SVM models were implemented in python sklearn 0.18.2 [26]. Default hyper parameters were used for RF models; the number of neighbours in KNN model was 9. The hyper parameters used for SVM model were C = 100 and ϵ= 0.5. ANN model was implemented in tensorflow 1.4.0 [27]. The ANN model contained six fully connected hidden layers and each layer contained 256 neurons with rectified linear unit activation [28]. For RF models, AAs were encoded as distinct integers. For other models, AAs were encoded as three numerical values using the schema in a relevant paper [29], as in Table S1.

Bacterial strains and culture conditions

To further evaluate the model performance on identifying resistance phenotypes, 62 clinical *S. pneumoniae* strains were collected from a local hospital. Pneumococci were cultivated in brain heart infusion broth (BD) or on trypticase soy agar (BD) plates containing 5% defibrinated sheep blood at 35°C with 5% CO_2 as described previously [30]. The MICs of the β-lactam antibiotics (CFX and AMO were used in this study) were determined by microdilution method according to CLSI guidelines (http://www.clsi.org/). Sequences of *pbp* genes were determined by Sanger sequencing. MIC thresholds for susceptibility and resistance for clinical isolates were determined based on current CLSI breakpoints (http://www.clsi.org/).

Construction of transformants

LMG87, S-186 and type strain: N were predicated as resistant strains, SMRU2896 as sensitive strain. *pbp1a*, *pbp2x* and *pbp2b* of *S. pneumoniae* LMG87, S-186, type strain: N and SMRU2896 were synthesized by GenScript (Nanjing, China). The synthesized genes were cloned into a plasmid pUC57 (Y14837.1), respectively, and the recombinant plasmids were transformed into *Escherichia coli* DH5α. The PCR products of *pbp1a* and *pbp2x* from three strains, LMG87, S-186, type strain: N (recombinant plasmids as amplification template) were assembled with PCR product of *S. pneumoniae* R6 *pbp1a* and *pbp2x* flanking region (2000 bp upstream and 2000 bp downstream of these genes), separately, using Gibson's one-step ISO assembly of overlapping double-stranded DNA (dsDNA) Kit (NEBuilder HiFi DNA Assembly Master Mix) [31]. The fusion fragments were used to transform *S. pneumoniae* R6, which is a descendant of the type 2 capsule S (smooth or encapsulated) clinical isolate D39 [32], and the transformants were selected on media containing 0.1 mg/L CFX. Genetic transformation was performed as previously described [7, 33]. In the 2nd round of transformation, the *pbp2b* genes and three strains were used as donor DNA and the transformats, $R6_{LMG87}$–1, $R6_{S\text{-}186}$-1 and $R6_{\text{type strain: N}}$-1, were receipts. Transformants were screened on a plate containing 2.5 mg/L AMO. To ensure functionality of synthetic *pbp1a* and *pbp2x* from SMRU2896, which was predicted as sensitive strain, genes were synthetized and cloned into an *E. coli-Streptococcus* shuttle plasmid pIB166 (conferring chloramphenicol resistance [Cmr]) [34]. The *pbp* genes were inserted downstream of a constitutive P23 promoter performed as described above [31]. The plasmids were transformed into R6. All constructs were confirmed by DNA sequencing and primers are listed in Table S2 in the supplemental material.

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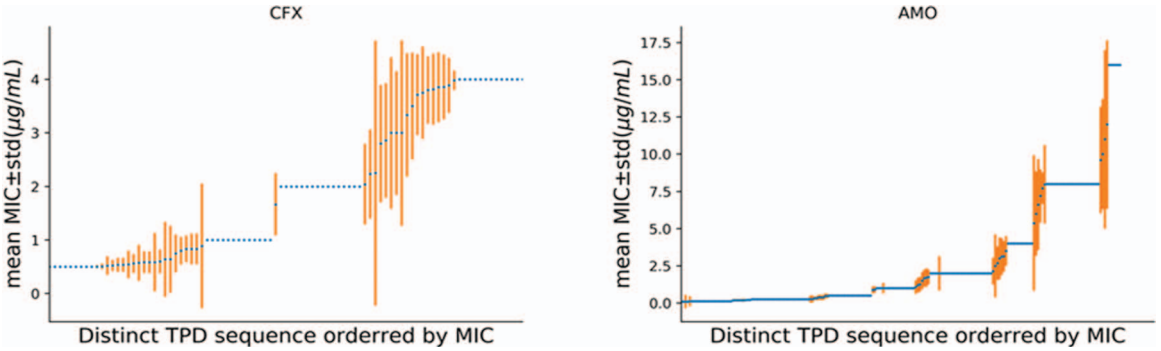


Figure 1. Mean MIC values of CFX and AMO to distinct TPD sequences. Vertical lines represent the standard deviation of MIC values of the same TPD sequence. TPD: Transpeptidase domain of three PBP was used to predict as input feature. SL models: Supervised learning models, RF: random forest model, KNN: K nearest neighbours model, SVM:support vector machine, ANN: Artificial neural network.

Predication of resistance phenotypes of clinical isolates

All data from Li et al. [35] were used as the training set to predict the MIC values of the 62 clinical *S. pneumoniae* strains. As the MIC values of CFX larger than 4 mg/L were truncated to 4 mg/L in training data, and the MIC values of AMO larger than 8 mg/L were truncated to 8 mg/L, the same truncation in training data was applied to predict the MICs of the 62 strains.

Analysis of *S. pneumoniae* genomes from NCBI database

The FASTA Nucleic Acid (fna) files of 8238 *S. pneumoniae* genomes were used to obtain TPD sequences and HVLs for the ML models described above to predict CFX or AMO resistant phenotypes. A total of 100 out of 8238 genomes failed to extract corresponding TPD sequences and were not included in this study. The Bioperl-1.5.2_100 built-in script Genbank2gff3.pl was used to convert the GenBank flatfile (GBFF) files to the Generic Feature Format Version 3 (GFF3) format, which was used as input of Roary v3.11.4 [36] to construct the core genomes of *S. pneumoniae* strains with 80% minimum percentage identity. A multi-FASTA alignment of the entire core genes were created by Roary v3.11.4 [36], which were then used to generate the maximum-likelihood core genome phylogenetic tree using FastTree 2.1.10 [37] with default parameters. The serotype and multilocus sequence type (ST) of 8138 *S. pneumoniae* strains were predicted by blast with MLST database and serotyping software (https://pubmlst.org/) [37, 38]. The eBURST v3 online software [39] and phyloviz v2.0 [40] were used to assign the clonal complex (CC) of CFX-nonsusceptible strain STs based on the stringent group definition of six out of seven shared alleles. Collection country and isolation source were obtained from GBFF files.

Results and discussion

Evaluation of four ML models to predicate the resistance phenotypes according to TPD sequences in *S. pneumoniae*

The strong correlation between the TPD AA sequences of PBP1a, PBP2b and PBP2x with the resistance phenotypes demonstrated by Li et al [20] is the foundation for our modelling study. We used their dataset to evaluate correlation between TPD sequences and resistance phenotypes and obtained similar results [20, 21]. However, we used a different model evaluation metric as shown in equation 3. MIC values were measured using a series of MIC concentrations, where every next concentration is doubled, and thus the MIC ranges were measured. Equation 3 handles this data inaccuracy by comparing the antibiotics resistance categories between experimentally measured and predicted values and catches the ordinal difference among categories [41].

$$M = 1 - \sum_{i=0}^{N} \frac{|c_i - c_{pi}|}{N} \quad (3)$$

Antibiotic resistance phenotypes are categorized into three groups: sensitive (0), intermediate (1) and resistant (2). In equation 3, N is the number of predicted strains; c_i is the true category of strain i; and c_{pi} is the predicted category of strain i. The penalties of incorrect predictions are shown in Table S3.

There are a few exceptions of the data in this study that strains with identical TPD sequences could result in different MIC values, as shown in Figure 1. The inconsistency might result from noise, growth environment, experimental error and other factors or addition genes that also contribute to MIC values. This phenomenon limits the maximum performance of models that predict MIC values from TPD sequences. To evaluate the extent of association between TPD sequences and antibiotic resistance, for a given labelled strain, the mode categories of all strains with the same TPD sequence were treated as predicted categories, then the performance score (equation 3) of predicted category was calculated. For CFX, the score was 0.991 and for AMO the score was 0.974, which represent the best performances among all models that predicts CFX or AMO resistant types from TPD AA sequences can reach with the available data, respectively. On the other hand, a dummy guess of any strain as sensitive will result in 0.867 and 0.870 performance scores for CFX and AMO, respectively. These are the minimum performances any model should achieve.

In addition to RF model used by Lee et al [21] we tested correlation between TPD sequences and antibiotic resistance phenotypes using three other SL models—KNN, SVM and ANN—to show the robustness against different models (Table 1). The performances of SL models were evaluated by cross-validation: the labelled data set was randomly split into 80% training set and 20% test set 100 times. For every split, models were trained using training set and evaluated using test set with equation 3. The final performance score was averaged from 100 experiments. Some errors in test data were unavoidable as the same sequence existed in training data but the resistant category is different in test and training data. Cross-validation experiments showed that for antibiotics CFX, 73.4±13.5%, and for AMO, 85.5±7.0%, errors were due to inconsistencies between MIC values in training and test data. The performances of the four models were equally well, and ANOVA test cannot differentiate

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Table 1. Performance scores of predicting antibiotics resistance according to TDP with SL models

Models	Antibiotics	
	AMO	CFX
RF	0.961 ± 0.007	0.985 ± 0.006
KNN	0.962 ± 0.008	0.984 ± 0.005
SVM	0.956 ± 0.007	0.985 ± 0.005
DNN	0.960 ± 0.009	0.985 ± 0.006

them (Table S4). However, RF model does not need to consider numerical encoding of AAs.

The model performance of CFX is better than AMO, probably because the percentage of positive (intermediate and resistant) strains in training data for CFX (14.8%) is higher than AMO (7.9%), and for the same sequences, resistance to AMO is more variant compared to CFX (Figure 1).

Highly variant AA loci are associated with antibiotic resistance

The above results showed that the model built with SL has a great potential for predicting antibiotic resistance phenotypes. However, there is a possibility that not all AA sequences of *pbp1a*, *pbp2b* and *pbp2x* are needed for the prediction since most of the AA sequences are conserved. The variance of each locus can be calculated from increasing amount of unlabelled data in public database like NCBI. *pbp1a*, *pbp2b* and *pbp2x* genes of 6748 *S. pneumoniae* strains were downloaded from NCBI database. For each AA loci, categorical variance [22] was calculated by comparing AAs of all strains at the same location, shown in Figure 2A. The probability densities of categorical variance were calculated and shown in Figure 2B. The AA loci can be separated into high-variance and low-variance categories. Separation boundaries were assumed to be the 1st local minimum of their density curves as shown in Figure 2B. The numbers of HVLs are shown in Table S5. It was hypothesized that these HVLs are associated with antibiotic resistance. The AA loci with high coefficient of determination (HCDs) (equation 2) to labelled MIC values were compared with HVLs. HCDs are the loci that statistically contribute mostly to the variance of MIC values. For CFX, which only relies on *pbp1a* and *pbp2x* genes, corresponding HVLs and HCDs overlap very well. For AMO, most HVLs and HCDs overlap in *pbp1a*, *pbp2b* and *pbp2x* genes, as shown in Figure S1. Thus, in general, HVLs are those loci contributing mostly to the variance of MIC values.

We next compared SL models built using whole TPD sequences and TPD HVLs as input features. The data were randomly split into 80% (3440) training set and 20% (860) test set. RF, KNN, SVM and ANN models based on HVLs achieved the same performance as these models based on TPDs. There is no loss of prediction power by reducing the model features from whole TPD sequences to HVLs (Figure 3). Thus, only these HVLs are required to predict antibiotic resistance.

Multi-fragments of AA sequences predict antibiotic resistance with equally good performance to full sequences

Application of current SL models for antibiotic resistance prediction rely on the full length sequence information of TPD sequences, which are time consuming and costly to acquire. The required sequencing length for model input might be shortened by exploiting the association between HVLs. Associated locus pairs help to reduce model features even without direct relationship [35]. Cramer's V is a statistical indicator of the association between two categorical variables, combining chi-squared test and sample size. Bias correction is applied when sample size is small to reduce overestimation of association strength [25]. Cramer's V with bias correction for 9591 possible pairs of 139 HVLs were calculated, and 2237 pairs had strong association (Cramer's V $\geq$ 0.7), as shown in Figure 4.

Association between HVLs implied that not all HVLs are needed to predict MIC values. Specifically, around 250 continuous AA sequences can be measured accurately in one 1st generation sequencing reaction [42]. As AA loci in *pbp1a*, *pbp2b* and *pbp2x* are strongly associated, it was suspected that one sequencing reaction for one gene would contain sufficient information for TPD sequences. To validate this hypothesis, AA loci fragment, 392–641 of *pbp1a*, 406–655 of *pbp2b*, and 337–586 of *pbp2x* were selected to predict the remaining HVLs, as they contain maximum cumulative categorical variance with fixed AA length 250 within corresponding genes. Sequences of 6748 strains from NCBI database were randomly split into 80% (5398) training set and 20% (1350) test set and trained a RF classification model using the training set (Table 2). On test data, the model predicted remaining HVLs with accuracy 99.14%, while a dummy guess of the most frequent AA resulted in only 84.49%

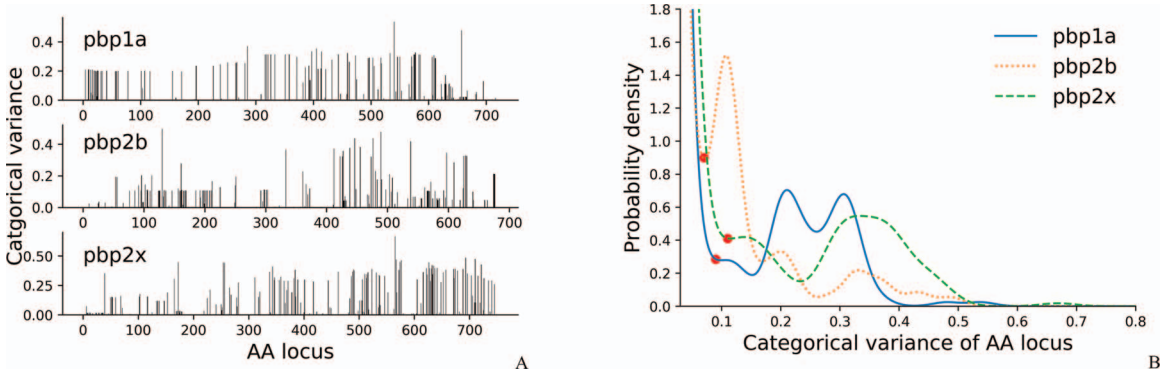


Figure 2. Categorical variance of pbp1a, pbp2b and pbp2x AA loci calculated from 6748 *S. pneumoniae* strains. Categorical variance is a statistical indicator of whether a specified locus is variant among different strains (**A**). Probability densities of categorical variance for pbp1a, pbp2b and pbp2x AA sequences. The 1st local minimum (red dots) of the probability density curve was assumed to separate loci into highly variant and conservative categories for each gene (**B**).

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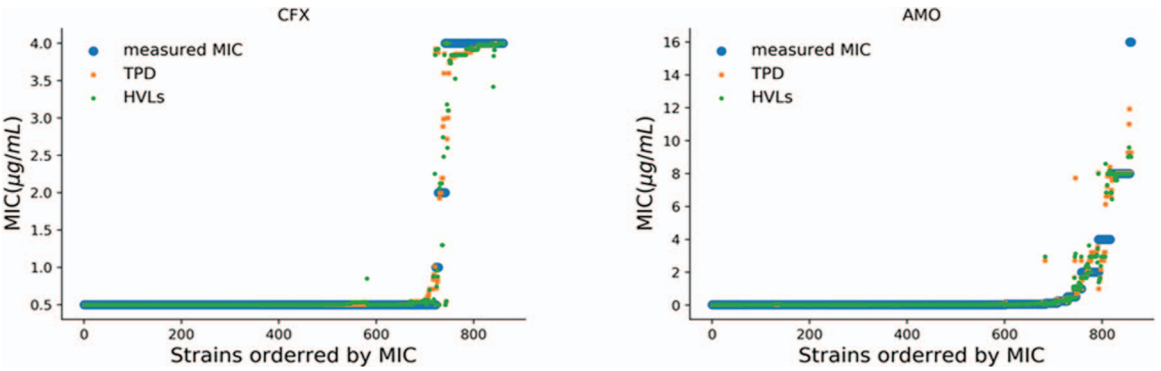


Figure 3. RF model predictions on test data set. A total of 4300 strains with measured MIC values were randomly split into 80% training set and 20% test set. RF models with different features were trained by training data and evaluated on the test data. TPD model used whole TPD sequences of pbp1a, pbp2b and pbp2x as model features and HVLs model used only HVLs as model features.

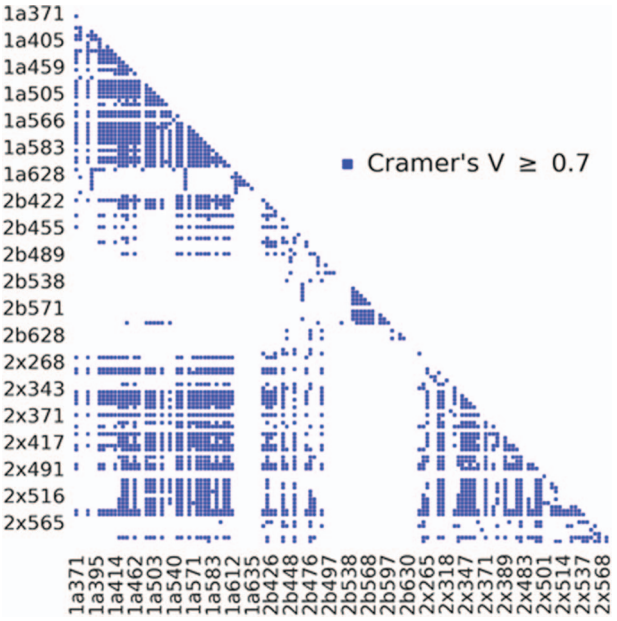


Figure 4. Cramer's V between HVLs. Blue block indicates the value of Cramer's V between two loci is greater than 0.7. Both row and column contain 139 HVLs. Loci were selectively presented as figure labels for visualization purpose.

accuracy. Thus, fragment of continuous 250 AA loci from each *pbp1a*, *pbp2b* and *pbp2x* genes contain most of the information of TPD sequences, and they were supposed to predict antibiotic resistance reasonably well. PBP2b and PBP2x fragments were used as model input features to predict AMO and CFX resistance, respectively. There is no compromise of model performance comparing to models using HVLs or TPD sequences as input features. ANOVA test cannot differentiate the performance of TPD sequence, HVLs and sequence fragment (Table S4, except CFX prediction by SVM model). The length of fragments *pbp2b* and *pbp2x* is 750 bp, which can be sequenced by Sanger sequencing in a single reaction. This means that only two reactions are needed to obtain the data for predicating the CFX and AMO resistance phenotypes. We designed the primers for fragments of *pbp2b* and *pbp2x* according to the conserved region, respectively (Table S6).

Experimental validation of the performance of the SL model

Sequences of *pbp2b* and *pbp2x* were extracted from *S. pneumoniae* genomes from NCBI database and their corresponding resistance phenotypes were predicated using our SL model. Four isolates were randomly selected where three of them were predicted as resistant to and one of them was predicted as susceptible to CFX and AMO (Table S7). These isolates contain diverse *pbp1a*, *pbp2x* and *pbp2b* and were isolated from different countries (Table S8). The genes of PBP were synthesized from these predicted resistant isolates and further transformed into the strain *S. pneumoniae* R6, which is susceptible to β-lactam. Transformants were selected in the presence of CFX and AMO, respectively. Transformants $R6_{LMG87}$-1, $R6_{S\text{-}186}$-1 and $R6_{\text{type strain: N}}$-1 not only increased the resistance to CFX by 512-1024-fold but also to 3rd-generation β-lactam, ceftriaxone by 64-256-fold compared to the R6 and increased the resistance to AMO by 8-fold (Table 3). Sequencing analysis showed that all of the transformants carried *pbp1a* and *pbp2x* genes of corresponding donor strain LMG87, S-186 and type strain: N. However, the AMO resistance levels of these transformants are still below the breakpoint. $R6_{LMG87}$-1, $R6_{S\text{-}186}$-1 and $R6_{\text{type strain: N}}$-1 were further transformed by *pbp2b* genes from corresponding strains, and the new transformants became resistant to AMO (Table 3). *pbp1a*, *pbp2x* and *pbp2b* from predicted susceptible strain SMRU2896 were cloned into shuttle plasmid pIB166 [36] and transformed into R6 as control. These *pbps* did not increase R6's resistance to CFX and AMO. These results validated that our SL model could provide the accurate prediction for resistance phenotypes according to PBP sequences.

In order to further evaluate the accuracy of the SL model, we predicated the MIC values of CFX and AMO of 62 clinical strains collected from the local hospital. At same time, the MICs were determined experimentally using microdilution methods according to CLSI (Table S9). For CFX, 60 out of 62 strains were predicted correctly (Table S9, Figure S2). One sensitive strain, A10 and one intermediate strain L310 were predicted as resistant strains. The TPD sequences A10 were present in training data, and its mean MIC value was 4.0 mg/L. Therefore, this is unavoidable error. The TPD sequences of L310 strain were identical to L400 strain, which is resistant in our collection. Thus, there may be other mechanisms contributing to L400's resistance phenotype. For AMO, 57 out of 62 strains were predicted correctly (Table S9, Figure S2). Intermediate strain A28 was predicted as

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Table 2. Performance scores of predicting antibiotics resistance with SL models

Models	AMO			CFX		
	HVLs	3slices	PBP2b	HVLs	3slices	PBP2x
RF	0.963 ± 0.005	0.962 ± 0.005	0.956 ± 0.007	0.984 ± 0.005	0.985 ± 0.005	0.986 ± 0.005
KNN	0.961 ± 0.007	0.960 ± 0.006	0.954 ± 0.009	0.985 ± 0.006	0.983 ± 0.006	0.984 ± 0.006
SVM	0.963 ± 0.006	0.957 ± 0.007	0.960 ± 0.006	0.986 ± 0.004	0.983 ± 0.005	0.984 ± 0.005
DNN	0.962 ± 0.006	0.961 ± 0.010	0.958 ± 0.010	0.985 ± 0.006	0.986 ± 0.006	0.985 ± 0.005

HVLs: Highly Variant Locis were used to predict as input feature. 3slices: 3 fragments captured from 3 PBP ends in series were used to predict as input feature. PBP2b: Slice from PBP2b was used to predict as input feature. PBP2x: Slice from PBP2x was used to predict as input feature.

Table 3. The MIC of R6 and transformants

Strains	Recipient	Donor	Transformed DNA	MIC (mg/L)		
				Cefuroxime	Ceftriaxone	Amoxicillin
R6	-	-	-	0.03125	0.015625	0.03125
$R6_{LMG87}$-1	R6	LMG87	*pbp1a*, *pbp2x*	32	4	0.25
$R6_{S\text{-}186}$-1	R6	S-186	*pbp1a*, *pbp2x*	32	2	0.25
$R6_{type\ strain:\ N}$-1	R6	type strain: N	*pbp1a*, *pbp2x*	16	1	0.25
$R6_{LMG87}$-2	$R6_{LMG87}$-1	LMG87	*pbp2b*	32	4	64
$R6_{S\text{-}186}$-2	$R6_{S\text{-}186}$-1	S-186	*pbp2b*	32	2	32
$R6_{type\ strain:\ N}$-2	$R6_{type\ strain:\ N}$-1	type strain: N	*pbp2b*	16	1	16
$R6_{SMRU2896}$	R6	SMRU2896	*pbp1a*, *pbp2x*	0.03125	0.015625	0.03125

resistant; resistant strain L377, intermediate strains M307 and Q16 were predicted as sensitive. The TPD sequences of A28 were also present in training data, and its mean MIC value was higher than 8 mg/L. The TPD sequences of strains L377, M307 and Q16 are very distant from existing strains in the training data, as in Figure S3. The error of these three strains might result from different distribution of training and test data, and accumulation of training data that cover more possible distributions might eliminate this kind of error [43]. The experimental validation of the performance of the SL model demonstrated that our model can predicate resistance phenotypes accurately for most of tested strains with limited prediction errors. The errors might be due to the limitation of models, training dataset and the non-PBP β-lactam resistance mechanism in *S. pneumonia*. [11].

Application of the SL model to predict β-lactam resistant *S. pneumoniae* in NCBI database

In the 8138 *S. pneumoniae* strains where genomes are available from NCBI database, 1550 strains were predicated as resistant strains against CFX, and 26 strains were predicated as CFX-intermediate. Among these 1576 CFX-nonsusceptible *S. pneumoniae* strains, 212 strains were predicated to be resistant to AMO, 571 be AMO-intermediate, and 793 strains be susceptible to AMO. Interestingly, only one strain is predicated to be AMO intermediate among 6562 strains that are predicted to be CFX susceptible (Table S10). A total of 23 serotypes were involved in the 1576 CFX-nonsusceptible strains, of which 11 serotypes (90 strains) were not covered by existing vaccines (Table S11) [44]. Serotype 35b is one of nonvaccine serotypes with 62 strains and 49% CFX-nonsusceptible rate. These strains were isolated from blood samples and nasopharyngeal swabs in the United States. This result was consistent with a previous report [45] indicating that the antibiotic resistance of *S. pneumoniae* serotype 35b strain, which was not covered by vaccine and caused both invasive and noninvasive disease, is emerging [45].

There are 989 STs in 8138 *S. pneumoniae* strains, and 243 strains did not have the corresponding ST. 1576 CFX-nonsusceptible strains distributed in 219 STs with a total of 2471 strains (Table S10). Among 2471 strains, all strains of 185 STs are predicated to be nonsusceptible, and more than 50% strains of 15 STs were nonsusceptible. The strains of these 200 STs with a high CFX nonsusceptible rate belong to 87 CCs, of which the predominant clones are CC236, CC90 and CC448.

Conclusion

It was extensively studied that the variations of PBPs have influenced β-lactam MICs in *S. pneumoniae* [11, 46]. Li *et al* proposed the PBP typing system that links the TPDs of PBP1a, PBP2b and PBP2x to β-lactam resistance phenotypes, and they constructed the predication model according to typing system [20]. Covariation of loci is often observed in AA sequence [47]. Direct coupling analysis showed large amounts of loci couplings exist between *pbp1a*, *pbp2b* and *pbp2x* genes in *S. pneumoniae* [48]. Therefore, in this study, we utilized the unlabeled data in public database in NCBI and demonstrated the highly variant AA loci are associated with antibiotic resistance. Association between AA loci was demonstrated by us as an approach for feature reduction using only HVLs or sequence fragments during building ML models. And finally, using only a fragment from *pbp2x* (750 bp) and a fragment from *pbp2b* (750 bp), we were able to predicate the CFX and AMO resistance well. This approach improved the practical application of molecular diagnostics to detect the β-lactam resistance in *S. pneumoniae*. Furthermore, we evaluated the precision of predication model by constructing the mutants containing the *pbps* from *S. pneumoniae* strains, of which genomes are available in NCBI database, and their phenotypes were predicated according to our model. We also tested our model by predicting resistance phenotypes of a local clinical strain collection. Both these approaches validated that our SL model could predicate the phenotype accurately. Besides, we applied our approach to ana-

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lyze the sequenced *S. pneumoniae* strains available in database, which provided association between resistance phenotype and serotype and STs of *S. pneumonia*. These results are helpful in understanding the worldwide epidemiology of *S. pneumonia*.

Key Points

- We apply supervised ML to predict *S. pneumoniae* antimicrobial susceptibility, combining the unlabeled data in public database in NCBI and published labeled data.
- We develop an approach using only short sequence to predicate antimicrobial susceptibility, which could be acquired by one Sanger sequencing reaction.
- We predicate the antimicrobial susceptibility of 8138 *S. pneumoniae* strains available in database using our approach.

Supplementary Data

Supplementary data are available online at https://academic.oup.com/bib.

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